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Time to Practice: Representing Text as a Vector
Graded Assignment • 15 min

Due Oct 23, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. If a document contains repeated occurrences of certain terms, which statistic should be used in the document vector representation to capture the number of times each term appears? 1 / 1 point

- Cosine Similarity
- Term Frequency (TF)
- Inverse Document Frequency (IDF)
- Document Frequency (DF)

Nice work

The Term Frequency (TF) representation records an integer value for each term that indicates how many times the term occurred in the document, which is suitable for capturing repeated occurrences.

2. What is the purpose of character normalization in text processing? 1 / 1 point

- Increasing text complexity
- Enhancing text readability
- Standardizing text format
- Reducing data storage needs

Nice work

Character normalization involves processing individual characters to convert them into a standardized format, such as converting all uppercase letters to lowercase, to reduce variability in the text.

3. A researcher is analyzing a collection of documents to determine a good set of keywords for each document. A good keyword is one that represents what's discussed frequently in the document, but is also a word that is not extremely common across all documents. What term weighting scheme should they use to find good keywords? 1 / 1 point

- Document Frequency (DF)
- TF-IDF weighting
- Inverse Document Frequency (IDF)
- Term Frequency (TF)

Nice work

TF-IDF term weight is used to capture both the frequency of a term in a document and its rarity across all documents. This makes it ideal for identifying unique terms that are important in individual documents.

4. A linguist is analyzing a corpus where variants of the word 'compute' such as 'computing' and 'computed' appear frequently. What text processing step should be applied to normalize these terms? 1 / 1 point

- Part-of-Speech Tagging
- Tokenization
- Stemming

Nice work

Stemming is a process that reduces words to their root form. Applying stemming to words like 'computing', 'computed', etc., will normalize them to 'compute', helping in treating different variants of the word as the same term, which is useful for analysis and search tasks.

5. In what scenario could the use of a simple binary bag-of-words model be justified over using a more sophisticated representation such as TF-IDF term weights? 1 / 1 point

- When the dataset is extremely large
- When the text data is non-English
- When term presence is more significant than frequency
- When computational resources are limited

Nice work

A binary bag-of-words model could be sufficient in scenarios where the mere presence of terms is more significant than the frequency of terms, such as in certain classification tasks where specific keywords are strong, important indicators of relevance or class membership, regardless of their frequency.

6. A team is developing a text processing pipeline for customer reviews. What step is crucial to ensure that terms like 'Email' and 'email' are treated the same? 1 / 1 point

- Tokenization
- Case conversion
- Stemming
- Stopword Removal

Nice work

Case conversion, a form of character normalization, ensures that words are treated uniformly regardless of their case, which is important for consistent text analysis.